

Laiba Fatima 22k-5195 Lab 7

```
ubuntu@ubuntu-virtual-machine: ~/ns-allinone-3.43/n...
sudo apt install csound
ubuntu@ubuntu-virtual-machine:~/ns-allinone-3.43$ cd ns-4.43
bash: cd: ns-4.43: No such file or directory
ubuntu@ubuntu-virtual-machine:~/ns-allinone-3.43$ cd ns-3.43/
ubuntu@ubuntu-virtual-machine:~/ns-allinone-3.43/ns-3.43$ ./ns3 run hello-
simulator
[0/2] Re-checking globbed directories...
ninja: no work to do.
Hello Simulator
ubuntu@ubuntu-virtual-machine:~/ns-allinone-3.43/ns-3.43$ ./ns3 run first
[0/2] Re-checking globbed directories...
ninja: no work to do.
At time +2s client sent 1024 bytes to 10.1.1.2 port 9
At time +2.00369s server received 1024 bytes from 10.1.1.1 port 49153
At time +2.00369s server sent 1024 bytes to 10.1.1.1 port 49153
At time +2.00737s client received 1024 bytes from 10.1.1.2 port 9
ubuntu@ubuntu-virtual-machine:~/ns-allinone-3.43/ns-3.43$ ./ns3 run second
[0/2] Re-checking globbed directories...
ninja: no work to do.
At time +2s client sent 1024 bytes to 10.1.2.4 port 9
At time +2.0118s server received 1024 bytes from 10.1.1.1 port 49153
At time +2.0118s server sent 1024 bytes to 10.1.1.1 port 49153
At time +2.02161s client received 1024 bytes from 10.1.2.4 port 9
ubuntu@ubuntu-virtual-machine:~/ns-allinone-3.43/ns-3.43$ SS
```

Task1

```
ubuntu@ubuntu-virtual-machine: ~/ns-allinone-3.43/ns-3.43
ubuntu@ubuntu-virtual-machine:~/ns-allinone-3.43$ ./ns3 run scratch/tcp-congestion
bash: ./ns3: No such file or directory
ubuntu@ubuntu-virtual-machine:~/ns-allinone-3.43$ cd ns-3.43
ubuntu@ubuntu-virtual-machine:~/ns-allinone-3.43/ns-3.43$ ./ns3 run scratch/tcp-congestion
- Using default output directory /home/ns-allinone-3.43/ns-3.43/build
- CCache is enabled.
- Proceeding without cmake-format
- find_external_library: SQLite3 was found.
- GSL was found.
- Precompiled headers were enabled.
- Processing src/antenna
- Standard library Bessel function has been found
- Processing src/aodv
- Processing src/applications
- Processing src/bridge
- Processing src/brite
- Skipping src/brite
- Processing src/buildings
- Processing src/click
- Skipping src/click
- Processing src/config-store
- Processing src/core
- Boost Units have been found.
- Processing src/csma
- Processing src/csma-layout
- Processing src/dsdv
- Processing src/dsr
```

```
ubuntu@ubuntu:~$ ./configure
-- Processing src/core
-- Boost Units have been found.
-- Processing src/csma
-- Processing src/csma-layout
-- Processing src/dsdv
-- Processing src/dsr
-- Processing src/energy
-- Processing src/fd-net-device
-- Checking for module 'libdpdk'
-- No package 'libdpdk' found
-- Processing src/flow-monitor
-- Processing src/internet
-- Processing src/internet-apps
-- Processing src/lr-wpan
-- Processing src/lte
-- Processing src/mesh
-- Processing src/mobility
-- Processing src/netanim
-- Processing src/network
-- Processing src/nix-vector-routing
-- Processing src/olsr
-- Processing src/openflow
-- Skipping src/openflow
-- Processing src/point-to-point
-- Processing src/point-to-point-layout
-- Processing src/propagation
```

```
ubuntu@ubuntu:~$ ./configure
-- Processing src/wlanx
-- ---- Summary of ns-3 settings:
build profile           : default
build directory         : /home/ins-allnone-3.43/ns-3.43/build
build with runtime asserts : ON
build with runtime logging : ON
build version embedding : OFF (not requested)
BRITE Integration      : OFF (Missing headers: "Brite.h" and missing libraries: "brite")
DES Metrics event collection : OFF (not requested)
OPDK NetDevice         : OFF (not requested)
Emulation FdNetDevice  : ON
Examples               : ON
File descriptor NetDevice : ON
GNU Scientific Library (GSL) : ON
GtkConfigStore         : ON
LibXml2 support        : ON
MPI Support            : OFF (not requested)
ns-3 Click Integration  : OFF (Missing headers: "simclick.h" and missing libraries: "nsclick click")
ns-3 OpenFlow Integration : OFF (Missing headers: "openflow.h" and missing libraries: "openflow")
Netmap emulation FdNetDevice : OFF (missing dependency)
PyViz visualizer       : OFF (Python Bindings are disabled)
Python Bindings        : OFF (not requested)
SQLite support         : ON
Eigen3 support         : ON
Tap Bridge             : ON
Tap FdNetDevice        : ON
Tests                  : ON

Modules configured to be built:
antenna          aodv          applications
bridge           buildings  config-store
core             csma       csma-layout
dsdv             energy    energy
```

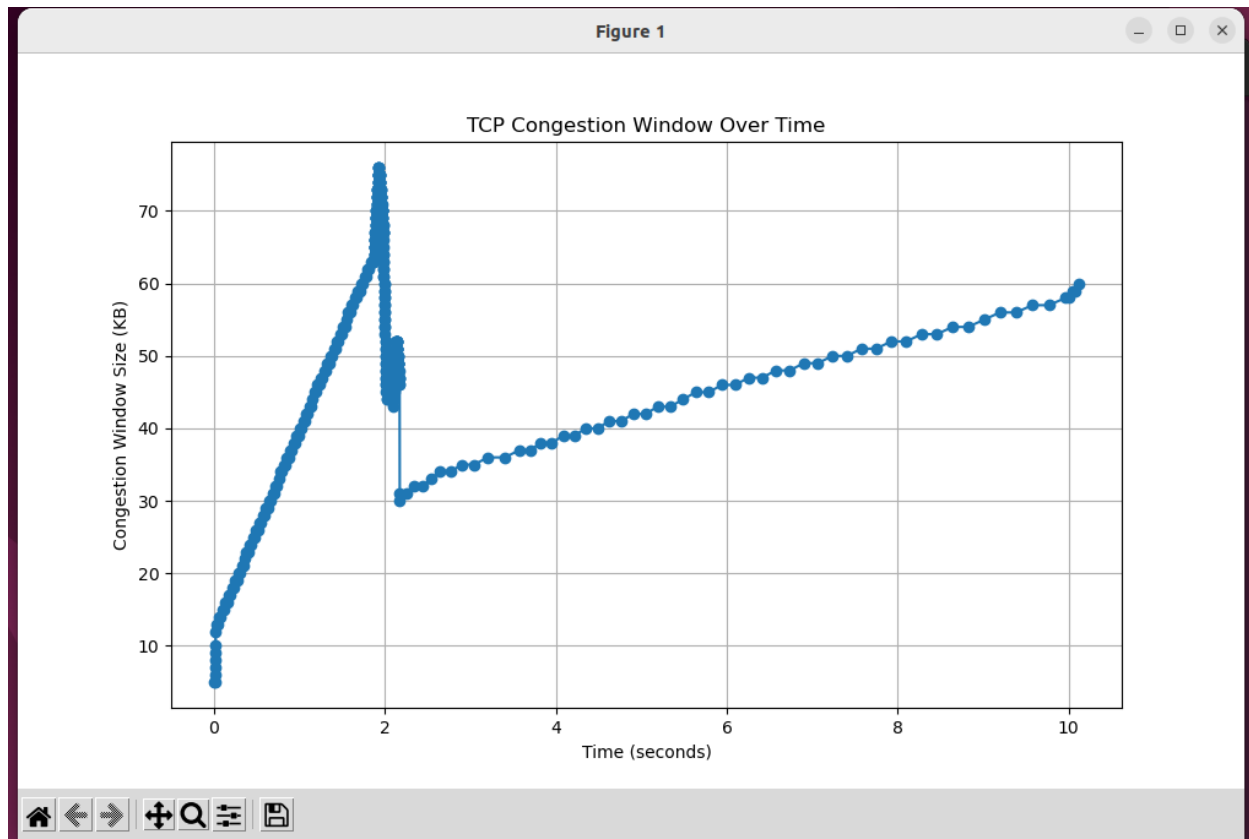
```
ubuntu@ubuntu-virt-uml-machine: ~/ns-allinone-3.43/ns-3.43
python Bindings : OFF (not requested)
Qlite support : ON
lgen3 support : ON
ap Bridge : ON
ap FdNetDevice : ON
ests : ON

Modules configured to be built:
antenna aodv applications
bridge buildings config-store
core csma csma-layout
dsdv dsr energy
d-net-device flow-monitor Internet
internet-apps lr-wpan lte
mesh mobility netanim
network nix-vector-routing olsr
point-to-point point-to-point-layout propagation
sdxlowpan spectrum stats
tap-bridge test topology-read
traffic-control uan virtual-net-device
wifi winax

Modules that cannot be built:
write click mpi
openflow visualizer

- Configuring done
- Generating done
```

```
Open [v] [f] cwnd.txt ~/ns-allinone-3.43/ns-3.43 Save [≡] [−] [□] [×]
tcp-congestion.cc x cwnd.txt x
1 0.0041856 5
2 0.0093024 5
3 0.0111904 6
4 0.0130784 7
5 0.0149664 8
6 0.0168544 9
7 0.0187424 10
8 0.0206304 12
9 0.0225184 13
10 0.0413984 13
11 0.0602784 14
12 0.0791584 14
13 0.0980384 15
14 0.116918 15
15 0.135798 16
16 0.154678 16
17 0.173558 17
18 0.192438 17
19 0.211318 18
20 0.230198 18
21 0.249078 19
22 0.267958 19
23 0.286838 20
24 0.305718 20
25 0.324598 21
26 0.343478 21
27 0.362358 22
28 0.381238 23
29 0.400118 23
30 0.418998 24
31 0.437878 24
32 0.456758 25
33 0.475638 25
34 0.494518 26
35 0.513398 26
36 0.532278 27
37 0.551158 27
38 0.570038 28
39 0.588918 28
40 0.607798 29
41 0.626678 29
42 0.645558 30
43 0.664438 30
44 0.683318 31
45 0.702198 31
46 0.721078 32
47 0.739958 32
48 0.758838 33
49 0.777718 33
50 0.796598 34
51 0.815478 34
52 0.834358 35
53 0.853238 35
54 0.872118 36
55 0.890998 36
56 0.909878 37
57 0.928758 37
58 0.947638 38
59 0.966518 38
60 0.985398 39
61 1.004278 39
62 1.023158 40
63 1.042038 40
64 1.060918 41
65 1.079798 41
66 1.098678 42
67 1.117558 42
68 1.136438 43
69 1.155318 43
70 1.174198 44
71 1.193078 44
72 1.211958 45
73 1.230838 45
74 1.249718 46
75 1.268598 46
76 1.287478 47
77 1.306358 47
78 1.325238 48
79 1.344118 48
80 1.362998 49
81 1.381878 49
82 1.400758 50
83 1.419638 50
84 1.438518 51
85 1.457398 51
86 1.476278 52
87 1.495158 52
88 1.514038 53
89 1.532918 53
90 1.551798 54
91 1.570678 54
92 1.589558 55
93 1.608438 55
94 1.627318 56
95 1.646198 56
96 1.665078 57
97 1.683958 57
98 1.702838 58
99 1.721718 58
100 1.740598 59
101 1.759478 59
102 1.778358 60
103 1.797238 60
104 1.816118 61
105 1.834998 61
106 1.853878 62
107 1.872758 62
108 1.891638 63
109 1.910518 63
110 1.929398 64
111 1.948278 64
112 1.967158 65
113 1.986038 65
114 2.004918 66
115 2.023798 66
116 2.042678 67
117 2.061558 67
118 2.080438 68
119 2.099318 68
120 2.118198 69
121 2.137078 69
122 2.155958 70
123 2.174838 70
124 2.193718 71
125 2.212598 71
126 2.231478 72
127 2.250358 72
128 2.269238 73
129 2.288118 73
130 2.306998 74
131 2.325878 74
132 2.344758 75
133 2.363638 75
134 2.382518 76
135 2.401398 76
136 2.420278 77
137 2.439158 77
138 2.458038 78
139 2.476918 78
140 2.495798 79
141 2.514678 79
142 2.533558 80
143 2.552438 80
144 2.571318 81
145 2.590198 81
146 2.609078 82
147 2.627958 82
148 2.646838 83
149 2.665718 83
150 2.684598 84
151 2.703478 84
152 2.722358 85
153 2.741238 85
154 2.760118 86
155 2.778998 86
156 2.797878 87
157 2.816758 87
158 2.835638 88
159 2.854518 88
160 2.873398 89
161 2.892278 89
162 2.911158 90
163 2.930038 90
164 2.948918 91
165 2.967798 91
166 2.986678 92
167 2.005558 92
168 2.024438 93
169 2.043318 93
170 2.062198 94
171 2.081078 94
172 2.100958 95
173 2.119838 95
174 2.138718 96
175 2.157598 96
176 2.176478 97
177 2.195358 97
178 2.214238 98
179 2.233118 98
180 2.251998 99
181 2.270878 99
182 2.289758 100
183 2.308638 100
184 2.327518 101
185 2.346398 101
186 2.365278 102
187 2.384158 102
188 2.403038 103
189 2.421918 103
190 2.440798 104
191 2.459678 104
192 2.478558 105
193 2.497438 105
194 2.516318 106
195 2.535198 106
196 2.554078 107
197 2.572958 107
198 2.591838 108
199 2.610718 108
200 2.629598 109
201 2.648478 109
202 2.667358 110
203 2.686238 110
204 2.705118 111
205 2.723998 111
206 2.742878 112
207 2.761758 112
208 2.780638 113
209 2.799518 113
210 2.818398 114
211 2.837278 114
212 2.856158 115
213 2.875038 115
214 2.893918 116
215 2.912798 116
216 2.931678 117
217 2.950558 117
218 2.969438 118
219 2.988318 118
220 3.007198 119
221 3.026078 119
222 3.044958 120
223 3.063838 120
224 3.082718 121
225 3.101598 121
226 3.120478 122
227 3.139358 122
228 3.158238 123
229 3.177118 123
230 3.195998 124
231 3.214878 124
232 3.233758 125
233 3.252638 125
234 3.271518 126
235 3.290398 126
236 3.309278 127
237 3.328158 127
238 3.347038 128
239 3.365918 128
240 3.384798 129
241 3.403678 129
242 3.422558 130
243 3.441438 130
244 3.460318 131
245 3.479198 131
246 3.498078 132
247 3.516958 132
248 3.535838 133
249 3.554718 133
250 3.573598 134
251 3.592478 134
252 3.611358 135
253 3.630238 135
254 3.649118 136
255 3.667998 136
256 3.686878 137
257 3.705758 137
258 3.724638 138
259 3.743518 138
260 3.762398 139
261 3.781278 139
262 3.800158 140
263 3.819038 140
264 3.837918 141
265 3.856798 141
266 3.875678 142
267 3.894558 142
268 3.913438 143
269 3.932318 143
270 3.951198 144
271 3.970078 144
272 3.988958 145
273 4.007838 145
274 4.026718 146
275 4.045598 146
276 4.064478 147
277 4.083358 147
278 4.102238 148
279 4.121118 148
280 4.139998 149
281 4.158878 149
282 4.177758 150
283 4.196638 150
284 4.215518 151
285 4.234398 151
286 4.253278 152
287 4.272158 152
288 4.291038 153
289 4.309918 153
290 4.328798 154
291 4.347678 154
292 4.366558 155
293 4.385438 155
294 4.404318 156
295 4.423198 156
296 4.442078 157
297 4.460958 157
298 4.479838 158
299 4.498718 158
300 4.517598 159
301 4.536478 159
302 4.555358 160
303 4.574238 160
304 4.593118 161
305 4.611998 161
306 4.630878 162
307 4.649758 162
308 4.668638 163
309 4.687518 163
310 4.706398 164
311 4.725278 164
312 4.744158 165
313 4.763038 165
314 4.781918 166
315 4.800798 166
316 4.819678 167
317 4.838558 167
318 4.857438 168
319 4.876318 168
320 4.895198 169
321 4.914078 169
322 4.932958 170
323 4.951838 170
324 4.970718 171
325 4.989598 171
326 5.008478 172
327 5.027358 172
328 5.046238 173
329 5.065118 173
330 5.083998 174
331 5.102878 174
332 5.121758 175
333 5.140638 175
334 5.159518 176
335 5.178398 176
336 5.197278 177
337 5.216158 177
338 5.235038 178
339 5.253918 178
340 5.272798 179
341 5.291678 179
342 5.310558 180
343 5.329438 180
344 5.348318 181
345 5.367198 181
346 5.386078 182
347 5.404958 182
348 5.423838 183
349 5.442718 183
350 5.461598 184
351 5.480478 184
352 5.499358 185
353 5.518238 185
354 5.537118 186
355 5.555998 186
356 5.574878 187
357 5.593758 187
358 5.612638 188
359 5.631518 188
360 5.650398 189
361 5.669278 189
362 5.688158 190
363 5.707038 190
364 5.725918 191
365 5.744798 191
366 5.763678 192
367 5.782558 192
368 5.801438 193
369 5.820318 193
370 5.839198 194
371 5.858078 194
372 5.876958 195
373 5.895838 195
374 5.914718 196
375 5.933598 196
376 5.952478 197
377 5.971358 197
378 5.990238 198
379 6.009118 198
380 6.027998 199
381 6.046878 199
382 6.065758 200
383 6.084638 200
384 6.103518 201
385 6.122398 201
386 6.141278 202
387 6.160158 202
388 6.179038 203
389 6.197918 203
390 6.216798 204
391 6.235678 204
392 6.254558 205
393 6.273438 205
394 6.292318 206
395 6.311198 206
396 6.330078 207
397 6.348958 207
398 6.367838 208
399 6.386718 208
400 6.405598 209
401 6.424478 209
402 6.443358 210
403 6.462238 210
404 6.481118 211
405 6.500038 211
406 6.518918 212
407 6.537798 212
408 6.556678 213
409 6.575558 213
410 6.594438 214
411 6.613318 214
412 6.632198 215
413 6.651078 215
414 6.669958 216
415 6.688838 216
416 6.707718 217
417 6.726598 217
418 6.745478 218
419 6.764358 218
420 6.783238 219
421 6.802118 219
422 6.820998 220
423 6.839878 220
424 6.858758 221
425 6.877638 221
426 6.896518 222
427 6.915398 222
428 6.934278 223
429 6.953158 223
430 6.972038 224
431 6.990918 224
432 7.009798 225
433 7.028678 225
434 7.047558 226
435 7.066438 226
436 7.085318 227
437 7.104198 227
438 7.123078 228
439 7.141958 228
440 7.160838 229
441 7.179718 229
442 7.198598 230
443 7.217478 230
444 7.236358 231
445 7.255238 231
446 7.274118 232
447 7.292998 232
448 7.311878 233
449 7.330758 233
450 7.349638 234
451 7.368518 234
452 7.387398 235
453 7.406278 235
454 7.425158 236
455 7.444038 236
456 7.462918 237
457 7.481798 237
458 7.500678 238
459 7.519558 238
460 7.538438 239
461 7.557318 239
462 7.576198 240
463 7.595078 240
464 7.613958 241
465 7.632838 241
466 7.651718 242
467 7.670598 242
468 7.689478 243
469 7.708358 243
470 7.727238 244
471 7.746118 244
472 7.764998 245
473 7.783878 245
474 7.802758 246
475 7.821638 246
476 7.840518 247
477 7.859398 247
478 7.878278 248
479 7.897158 248
480 7.916038 249
481 7.934918 249
482 7.953798 250
483 7.972678 250
484 7.991558 251
485 8.010438 251
486 8.029318 252
487 8.048198 252
488 8.067078 253
489 8.085958 253
490 8.104838 254
491 8.123718 254
492 8.142598 255
493 8.161478 255
494 8.180358 256
495 8.199238 256
496 8.218118 257
497 8.236998 257
498 8.255878 258
499 8.274758 258
500 8.293638 259
501 8.312518 259
502 8.331398 260
503 8.350278 260
504 8.369158 261
505 8.388038 261
506 8.406918 262
507 8.425798 262
508 8.444678 263
509 8.463558 263
510 8.482438 264
511 8.501318 264
512 8.520198 265
513 8.539078 265
514 8.557958 266
515 8.576838 266
516 8.595718 267
517 8.614598 267
518 8.633478 268
519 8.652358 268
520 8.671238 269
521 8.690118 269
522 8.708998 270
523 8.727878 270
524 8.746758 271
525 8.765638 271
526 8.784518 272
527 8.803398 272
528 8.822278 273
529 8.841158 273
530 8.860038 274
531 8.878918 274
532 8.897798 275
533 8.916678 275
534 8.935558 276
535 8.954438 276
536 8.973318 277
537 8.992198 277
538 9.011078 278
539 9.029958 278
540 9.048838 279
541 9.067718 279
542 9.086598 280
543 9.105478 280
544 9.124358 281
545 9.143238 281
546 9.162118 282
547 9.180998 282
548 9.199878 283
549 9.218758 283
550 9.237638 284
551 9.256518 284
552 9.275398 285
553 9.294278 285
554 9.313158 286
555 9.332038 286
556 9.350918 287
557 9.369798 287
558 9.388678 288
559 9.407558 288
560 9.426438 289
561 9.445318 289
562 9.464198 290
563 9.483078 290
564 9.501958 291
565 9.520838 291
566 9.539718 292
567 9.558598 292
568 9.577478 293
569 9.596358 293
570 9.615238 294
571 9.634118 294
572 9.652998 295
573 9.671878 295
574 9.690758 296
575 9.709638 296
576 9.728518 297
577 9.747398 297
578 9.766278 298
579 9.785158 298
580 9.804038 299
581 9.822918 299
582 9.841798 300
583 9.860678 300
584 9.879558 301
585 9.898438 301
586 9.917318 302
587 9.936198 302
588 9.955078 303
589 9.973958 303
590 9.992838 304
591 10.011718 304
592 10.030598 305
593 10.049478 305
594 10.068358 306
595 10.087238 306
596 10.106118 307
597 10.124998 307
598 10.143878 308
599 10.162758 308
600 10.181638 309
601 10.200518 309
602 10.219398 310
603 10.238278 310
604 10.257158 311
605 10.276038 311
606 10.294918 312
607 10.313798 312
608 10.332678 313
609 10.351558 313
610 10.370438 314
611 10.389318 314
612 10.408198 315
613 10.427078 315
614 10.445958 316
615 10.464838 316
616 10.483718 317
617 10.502598 317
618 10.521478 318
619 10.540358 318
620 10.559238 319
621 10.578118 319
622 10.596998 320
623 10.615878 320
624 10.634758 321
625 10.653638 321
626 10.672518 322
627 10.691398 322
628 10.710278 323
629 10.729158 323
630 10.748038 324
631 10.766918 324
632 10.785798 325
633 10.804678 325
634 10.823558 326
635 10.842438 326
636 10.861318 327
637 10.880198 327
638 10.899078 328
639 10.917958 328
640 10.936838 329
641 10.955718 329
642 10.974598 330
643 10.993478 330
644 11.012358 331
645 11.031238 331
646 11.050118 332
647 11.068998 332
648 11.087878 333
649 11.106758 333
650 11.125638 334
651 11.144518 334
652 11.163398 335
653 11.182278 335
654 11.201158 336
655 11.220038 336
656 11.238918 337
657 11.257798 337
658 11.276678 338
659 11.295558 338
660 11.314438 339
661 11.333318 339
662 11.352198 340
663 11.371078 340
664 11.389958 341
665 11.408838 341
666 11.427718 342
667 11.446598 342
668 11.465478 343
669 11.484358 343
670 11.503238 344
671 11.522118 344
672 11.540998 345
673 11.559878 345
674 11.578758 346
675 11.597638 346
676 11.616518 347
677 11.635398 347
678 11.654278 348
679 11.673158 348
680 11.692038 349
681 11.710918 349
682 11.729798 350
683 11.748678 350
684 11.767558 351
685 11.786438 351
686 11.805318 352
687 11.824198 352
688 11.843078 353
689 11.861958 353
690 11.880838 354
691 11.899718 354
692 11.918598 355
693 11.937478 355
694 11.956358 356
695 11.975238 356
696 11.994118 357
697 12.012998 357
698 12.031878 358
699 12.050758 358
700 12.069638 359
701 12.088518 359
702 12.107398 360
703 12.126278 360
704 12.145158 361
705 12.164038 361
706 12.182918 362
707 12.201798 362
708 12.220678 363
709 12.239558 363
710 12.258438 364
711 12.277318 364
712 12.296198 365
713 12.315078 365
714 12.333958 366
715 12.352838 366
716 12.371718 367
717 12.390598 367
718 12.409478 368
719 12.428358 368
720 12.447238
```



Plot cwnd.py

```
import matplotlib.pyplot as plt
# Lists to store time and congestion window size
time = []
cwnd = []
# Read the cwnd.txt file
with open("cwnd.txt", "r") as file:
    for line in file:
        parts = line.strip().split("\t")
        if len(parts) == 2:
            time.append(float(parts[0]))
            cwnd.append(float(parts[1]))
# Plotting
plt.figure(figsize=(10, 6))
plt.plot(time, cwnd, marker='o')
plt.title('TCP Congestion Window Over Time')
plt.xlabel('Time (seconds)')
plt.ylabel('Congestion Window Size (KB)')
plt.grid(True)
plt.show()
```



```
1 import matplotlib.pyplot as plt
2
3 # Lists to store time and congestion window size
4 time = []
5 cwnd = []
6
7 # Read the cwnd.txt file
8 with open("cwnd.txt", "r") as file:
9     for line in file:
10         parts = line.strip().split("\t")
11         if len(parts) == 2:
12             time.append(float(parts[0]))
13             cwnd.append(float(parts[1]))
14
15 # Plotting
16 plt.figure(figsize=(10, 6))
17 plt.plot(time, cwnd, marker='o')
18 plt.title('TCP Congestion Window Over Time')
19 plt.xlabel('Time (seconds)')
20 plt.ylabel('Congestion Window Size (KB)')
21 plt.grid(True)
22 plt.show()
23
```

Tcp-congestion.cc

```
#include "ns3/core-module.h"
#include "ns3/network-module.h"
#include "ns3/internet-module.h"
#include "ns3/point-to-point-module.h"
#include "ns3/applications-module.h"
#include "ns3/flow-monitor-module.h"
using namespace ns3;
NS_LOG_COMPONENT_DEFINE ("TcpCongestionControlExample");
// This function writes (time, cwnd) to file
void
CwndChange (Ptr<OutputStreamWrapper> stream, uint32_t oldCwnd, uint32_t newCwnd)
{
    *stream->GetStream () << Simulator::Now ().GetSeconds () << "\t" << newCwnd / 1024 <<
    std::endl;
}
// This function connects to the BulkSendApplication's socket *after* it starts
void
TraceCwnd (Ptr<Application> app, Ptr<OutputStreamWrapper> stream)
{
    Ptr<BulkSendApplication> bulkSendApp = DynamicCast<BulkSendApplication> (app);
    Ptr<Socket> sock = bulkSendApp->GetSocket ();
    if (sock)
    {
        sock->TraceConnectWithoutContext ("CongestionWindow", MakeBoundCallback
        (&CwndChange, stream));
    }
}
```

```

int
main (int argc, char *argv[])
{
    Time::SetResolution (Time::NS);
    NodeContainer nodes;
    nodes.Create (2);
    PointToPointHelper pointToPoint;
    pointToPoint.SetDeviceAttribute ("DataRate", StringValue ("5Mbps"));
    pointToPoint.SetChannelAttribute ("Delay", StringValue ("2ms"));
    NetDeviceContainer devices;
    devices = pointToPoint.Install (nodes);
    InternetStackHelper stack;
    stack.Install (nodes);
    Ipv4AddressHelper address;
    address.SetBase ("10.1.1.0", "255.255.255.0");
    Ipv4InterfaceContainer interfaces = address.Assign (devices);
    uint16_t port = 9; // Discard port
    BulkSendHelper source ("ns3::TcpSocketFactory",
                           InetSocketAddress (interfaces.GetAddress (1), port));
    source.SetAttribute ("MaxBytes", UIntegerValue (0));
    ApplicationContainer sourceApps = source.Install (nodes.Get (0));
    sourceApps.Start (Seconds (0.0));
    sourceApps.Stop (Seconds (10.0));
    PacketSinkHelper sink ("ns3::TcpSocketFactory",
                           InetSocketAddress (Ipv4Address::GetAny (), port));
    ApplicationContainer sinkApps = sink.Install (nodes.Get (1));
    sinkApps.Start (Seconds (0.0));
    sinkApps.Stop (Seconds (10.0));
    // Enable Congestion Window Tracing
    AsciiTraceHelper asciiTraceHelper;
    Ptr<OutputStreamWrapper> stream = asciiTraceHelper.CreateFileStream ("cwnd.txt");
    // Schedule tracing after application starts
    Simulator::Schedule (Seconds (0.001), &TraceCwnd, sourceApps.Get(0), stream);
    Simulator::Run ();
    Simulator::Destroy ();
    return 0;
}

```

Task2

Bandwidth_analysis.cc

```

#include "ns3/core-module.h"
#include "ns3/network-module.h"
#include "ns3/internet-module.h"
#include "ns3/point-to-point-module.h"
#include "ns3/applications-module.h"
#include "ns3/flow-monitor-module.h"

```

```

using namespace ns3;

NS_LOG_COMPONENT_DEFINE ("BandwidthAnalysis");

void RunSimulation (std::string bandwidth)
{
    // Create 2 nodes
    NodeContainer nodes;
    nodes.Create (2);

    // Configure point-to-point link
    PointToPointHelper pointToPoint;
    pointToPoint.SetDeviceAttribute ("DataRate", StringValue (bandwidth));
    pointToPoint.SetChannelAttribute ("Delay", StringValue ("2ms"));

    NetDeviceContainer devices = pointToPoint.Install (nodes);

    // Install Internet stack
    InternetStackHelper stack;
    stack.Install (nodes);

    // Assign IP addresses
    Ipv4AddressHelper address;
    address.SetBase ("10.1.1.0", "255.255.255.0");
    Ipv4InterfaceContainer interfaces = address.Assign (devices);

    // Set up UDP traffic
    uint16_t port = 9;
    UdpServerHelper server (port);
    ApplicationContainer serverApp = server.Install (nodes.Get (1));
    serverApp.Start (Seconds (1.0));
    serverApp.Stop (Seconds (10.0));

    UdpClientHelper client (interfaces.GetAddress (1), port);
    client.SetAttribute ("MaxPackets", UIntegerValue (32000));
    client.SetAttribute ("Interval", TimeValue (Milliseconds (10))); // 10 ms between packets
    client.SetAttribute ("PacketSize", UIntegerValue (1024));

    ApplicationContainer clientApp = client.Install (nodes.Get (0));
    clientApp.Start (Seconds (2.0));
    clientApp.Stop (Seconds (10.0));

    // Install FlowMonitor
    FlowMonitorHelper flowHelper;
    Ptr<FlowMonitor> monitor = flowHelper.InstallAll ();

    // Run Simulation
    Simulator::Stop (Seconds (11.0));
    Simulator::Run ();
}

```



```

// Flow Monitor Results
monitor->CheckForLostPackets ();
Ptr<Ipv4FlowClassifier> classifier = DynamicCast<Ipv4FlowClassifier>
(flowHelper.GetClassifier ());
std::map<FlowId, FlowMonitor::FlowStats> stats = monitor->GetFlowStats ();

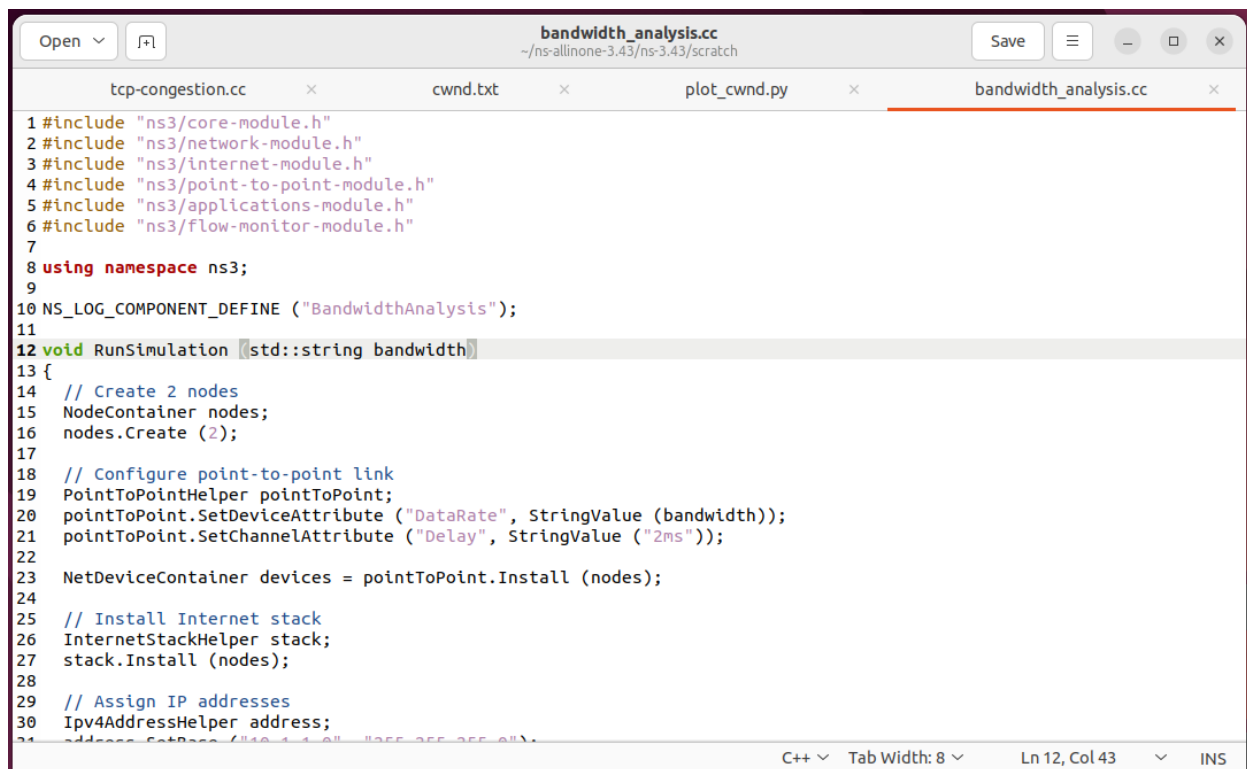
std::cout << "\nResults for Bandwidth = " << bandwidth << std::endl;
for (auto& flow : stats)
{
    Ipv4FlowClassifier::FiveTuple t = classifier->FindFlow (flow.first);
    if (t.destinationPort == port)
    {
        double throughput = flow.second.rxBytes * 8.0 /
(flow.second.timeLastRxPacket.GetSeconds () - flow.second.timeFirstTxPacket.GetSeconds ())
/ 1024 / 1024; // Mbps
        std::cout << "Flow ID: " << flow.first << ", Src Addr: " << t.sourceAddress << ", Dst Addr: "
<< t.destinationAddress << std::endl;
        std::cout << " Throughput: " << throughput << " Mbps" << std::endl;
        std::cout << " Packet Loss Ratio: " << (double)(flow.second.txPackets -
flow.second.rxPackets) / flow.second.txPackets << std::endl;
    }
}

Simulator::Destroy ();
}

int main (int argc, char *argv[])
{
    // Run for different bandwidths
    RunSimulation ("1Mbps");
    RunSimulation ("5Mbps");
    RunSimulation ("10Mbps");

    return 0;
}

```

```
bandwidth_analysis.cc
~/ns-allinone-3.43/ns-3.43/scratch

tcp-congestion.cc × cwnd.txt × plot_cwnd.py × bandwidth_analysis.cc ×

1 #include "ns3/core-module.h"
2 #include "ns3/network-module.h"
3 #include "ns3/internet-module.h"
4 #include "ns3/point-to-point-module.h"
5 #include "ns3/applications-module.h"
6 #include "ns3/flow-monitor-module.h"
7
8 using namespace ns3;
9
10 NS_LOG_COMPONENT_DEFINE ("BandwidthAnalysis");
11
12 void RunSimulation (std::string bandwidth)
13 {
14     // Create 2 nodes
15     NodeContainer nodes;
16     nodes.Create (2);
17
18     // Configure point-to-point link
19     PointToPointHelper pointToPoint;
20     pointToPoint.SetDeviceAttribute ("DataRate", StringValue (bandwidth));
21     pointToPoint.SetChannelAttribute ("Delay", StringValue ("2ms"));
22
23     NetDeviceContainer devices = pointToPoint.Install (nodes);
24
25     // Install Internet stack
26     InternetStackHelper stack;
27     stack.Install (nodes);
28
29     // Assign IP addresses
30     Ipv4AddressHelper address;
31     address.SetBase ("10.1.1.0", "255.255.255.0");
```

```
ubuntu@ubuntu-virtual-machine: ~/ns-allinone-3.43/ns-3.43$ ./ns3 run scratch/bandwidth_analysis
-- Using default output directory /home/ns-allinone-3.43/ns-3.43/build
-- CCache is enabled.
-- Proceeding without cmake-format
-- find_external_library: SQLite3 was found.
-- GSL was found.
-- Precompiled headers were enabled.
-- Processing src/antenna
-- Standard library Bessel function has been found
-- Processing src/andv
-- Processing src/applications
-- Processing src/bridge
-- Processing src/brite
-- Skipping src/brite
-- Processing src/buildings
-- Processing src/click
-- Skipping src/click
-- Processing src/config-store
-- Processing src/core
-- Boost Units have been found.
-- Processing src/csma
-- Processing src/csma-layout
-- Processing src/dsdv
-- Processing src/dsr
-- Processing src/energy
-- Processing src/fd-net-device
-- Checking for module 'libdpdk'
-- No package 'libdpdk' found
-- Processing src/flow-monitor
-- Processing src/internet
-- Processing src/internet-apps
-- Processing src/lr-wpan
-- Processing src/lte
-- Processing src/mesh
-- Processing src/mobility
```

```
-- Processing src/network
-- Processing src/nix-vector-routing
-- Processing src/olsr
-- Processing src/openflow
-- Skipping src/openflow
-- Processing src/point-to-point
-- Processing src/point-to-point-layout
-- Processing src/propagation
-- Processing src/sixlowpan
-- Processing src/spectrum
-- Processing src/stats
-- Processing src/tap-bridge
-- Processing src/test
-- Processing src/topology-read
-- Processing src/traffic-control
-- Processing src/uan
-- Processing src/virtual-net-device
-- Processing src/wifi
-- Processing src/wimax
-- ---- Summary of ns-3 settings:
Build profile           : default
Build directory         : /home/ns-allinone-3.43/ns-3.43/build
Build with runtime asserts : ON
Build with runtime logging : ON
Build version embedding : OFF (not requested)
BRITe Integration       : OFF (Missing headers: "Brite.h" and missing libraries: "brite")
DES Metrics event collection : OFF (not requested)
OPDK NetDevice          : OFF (not requested)
Emulation FdNetDevice   : ON
Examples                : ON
File descriptor NetDevice : ON
GNU Scientific Library (GSL) : ON
GtkConfigStore          : ON
LibXml2 support         : ON
MPI Support             : OFF (not requested)
```

```
GtkConfigStore          : ON
LibXml2 support         : ON
MPI Support             : OFF (not requested)
ns-3 Click Integration  : OFF (Missing headers: "simclick.h" and missing libraries: "nsclick cli
ck")
ns-3 OpenFlow Integration : OFF (Missing headers: "openflow.h" and missing libraries: "openflow")
Netmap emulation FdNetDevice : OFF (missing dependency)
PyViz visualizer        : OFF (Python Bindings are disabled)
Python Bindings         : OFF (not requested)
SQLite support          : ON
Eigen3 support          : ON
Tap Bridge              : ON
Tap FdNetDevice         : ON
Tests                   : ON
```

Modules configured to be built:

antenna	aodv	applications
bridge	buildings	config-store
core	csma	csma-layout
dsdv	dsr	energy
fd-net-device	flow-monitor	internet
internet-apps	lr-wpan	lte
mesh	mobility	netanim
network	nix-vector-routing	olsr
point-to-point	point-to-point-layout	propagation
sixlowpan	spectrum	stats
tap-bridge	test	topology-read
traffic-control	uan	virtual-net-device
wifi	wimax	

Modules that cannot be built:

brite	click	mpi
openflow	visualizer	

```

-- Processing src/network
internet-apps      lr-wpan           lte
mesh               mobility          netanim
network            nix-vector-routing  olsr
point-to-point     point-to-point-layout  propagation
sixlowpan          spectrum          stats
tap-bridge         test              topology-read
traffic-control    uan              virtual-net-device
wifi              winax

Modules that cannot be built:
brlte              click              npl
openflow          visualizer

-- Configuring done
-- Generating done
-- Build files have been written to: /home//ns-allinone-3.43/ns-3.43/cmake-cache
[0/2] Re-checking globbed directories...
[2/2] Linking CXX executable ../build/scratch/ns3.43-bandwidth_analysis-default

Results for Bandwidth = 1Mbps
Flow ID: 1, Src Addr: 10.1.1.1, Dst Addr: 10.1.1.2
Throughput: 0.802569 Mbps
Packet Loss Ratio: 0

Results for Bandwidth = 5Mbps
Flow ID: 1, Src Addr: 10.1.1.1, Dst Addr: 10.1.1.2
Throughput: 0.803246 Mbps
Packet Loss Ratio: 0

Results for Bandwidth = 10Mbps
Flow ID: 1, Src Addr: 10.1.1.1, Dst Addr: 10.1.1.2
Throughput: 0.803331 Mbps
Packet Loss Ratio: 0

```

Task3

Buffer_management.cc

```

#include "ns3/core-module.h"
#include "ns3/network-module.h"
#include "ns3/internet-module.h"
#include "ns3/point-to-point-module.h"
#include "ns3/applications-module.h"
#include "ns3/flow-monitor-module.h"

```

```
using namespace ns3;
```

```
NS_LOG_COMPONENT_DEFINE ("BufferManagementExample");
```

```
void RunSimulation (uint32_t queueSize)
```

```
{
    NodeContainer nodes;
    nodes.Create (2);
```

```
    PointToPointHelper pointToPoint;
    pointToPoint.SetDeviceAttribute ("DataRate", StringValue ("2Mbps"));
    pointToPoint.SetChannelAttribute ("Delay", StringValue ("2ms"));
```

```
    pointToPoint.SetQueue ("ns3::DropTailQueue",
```

```

        "MaxSize", QueueSizeValue (QueueSize (QueueSizeUnit::PACKETS,
queueSize)));

NetDeviceContainer devices = pointToPoint.Install (nodes);

InternetStackHelper stack;
stack.Install (nodes);

Ipv4AddressHelper address;
address.SetBase ("10.1.1.0", "255.255.255.0");
Ipv4InterfaceContainer interfaces = address.Assign (devices);

uint16_t port = 9;
OnOffHelper onoff ("ns3::UdpSocketFactory",
    InetSocketAddress (interfaces.GetAddress (1), port));
onoff.SetConstantRate (DataRate ("5Mbps")); // Higher than link capacity
onoff.SetAttribute ("PacketSize", UIntegerValue (1024));

ApplicationContainer app = onoff.Install (nodes.Get (0));
app.Start (Seconds (1.0));
app.Stop (Seconds (5.0));

PacketSinkHelper sink ("ns3::UdpSocketFactory",
    InetSocketAddress (Ipv4Address::GetAny (), port));
ApplicationContainer sinkApp = sink.Install (nodes.Get (1));
sinkApp.Start (Seconds (0.0));
sinkApp.Stop (Seconds (5.0));

FlowMonitorHelper flowmon;
Ptr<FlowMonitor> monitor = flowmon.InstallAll ();

Simulator::Stop (Seconds (6.0));
Simulator::Run ();

monitor->CheckForLostPackets ();

Ptr<Ipv4FlowClassifier> classifier = DynamicCast<Ipv4FlowClassifier> (flowmon.GetClassifier
());
std::map<FlowId, FlowMonitor::FlowStats> stats = monitor->GetFlowStats ();

for (auto it = stats.begin (); it != stats.end (); ++it)
{
    Ipv4FlowClassifier::FiveTuple t = classifier->FindFlow (it->first);
    if (t.destinationPort == port)
    {
        double packetLossRatio = (double)(it->second.lostPackets) / (it->second.txPackets);
        std::cout << "Queue Size = " << queueSize
            << " packets --> Packet Loss Ratio = "
            << packetLossRatio * 100 << " %" << std::endl;
    }
}

```

```

    }
}

Simulator::Destroy ();
}

int main (int argc, char *argv[])
{
    NS_LOG_UNCOND ("Starting Buffer Management Simulation...");

    RunSimulation (5); // Queue Size 5 packets
    RunSimulation (10); // Queue Size 10 packets
    RunSimulation (50); // Queue Size 50 packets

    return 0;
}

```



```

buffer_management.cc
~/ns-allinone-3.43/ns-3.43/scratch

bandwidth_analysis.cc x buffer_management.cc x

1 #include "ns3/core-module.h"
2 #include "ns3/network-module.h"
3 #include "ns3/internet-module.h"
4 #include "ns3/point-to-point-module.h"
5 #include "ns3/applications-module.h"
6 #include "ns3/flow-monitor-module.h"
7
8 using namespace ns3;
9
10 NS_LOG_COMPONENT_DEFINE ("BufferManagementExample");
11
12 void RunSimulation (uint32_t queueSize)
13 {
14     NodeContainer nodes;
15     nodes.Create (2);
16
17     PointToPointHelper pointToPoint;
18     pointToPoint.SetDeviceAttribute ("DataRate", StringValue ("2Mbps"));
19     pointToPoint.SetChannelAttribute ("Delay", StringValue ("2ms"));
20
21     // Fix the queue size initialization
22     pointToPoint.SetQueue ("ns3::DropTailQueue",
23         "MaxSize", QueueSizeValue (QueueSize (QueueSizeUnit::PACKETS, queueSize)));
24
25     NetDeviceContainer devices = pointToPoint.Install (nodes);
26
27     InternetStackHelper stack;
28     stack.Install (nodes);
29
30     Ipv4AddressHelper address;
31     address.SetBase ("10.1.1.0", "255.255.255.0");

```

```

ubuntu@ubuntu-virtual-machine:~/ns-allinone-3.43/ns-3.43$ ./ns3 run scratch/buffer_management
[0/2] Re-checking globbed directories...
[2/2] Linking CXX executable ../build/scratch/ns3.43-buffer_management-default
Starting Buffer Management Simulation...
Queue Size = 5 packets --> Packet Loss Ratio = 26.0549 %
Queue Size = 10 packets --> Packet Loss Ratio = 25.891 %
Queue Size = 50 packets --> Packet Loss Ratio = 24.8259 %

```

Task4

Bottleneck analysis.cc

```

#include "ns3/core-module.h"
#include "ns3/network-module.h"
#include "ns3/internet-module.h"
#include "ns3/point-to-point-module.h"
#include "ns3/applications-module.h"
#include "ns3/flow-monitor-module.h"

using namespace ns3;

NS_LOG_COMPONENT_DEFINE ("BottleneckAnalysisExample");

void RunSimulation (uint32_t trafficRate)
{
    std::cout << "\nRunning Simulation for Traffic Rate = " << trafficRate << " Mbps...\n";

    NodeContainer senders, receivers, routers;
    senders.Create (4);
    receivers.Create (4);
    routers.Create (2);

    // Set up point-to-point links between senders and routers
    PointToPointHelper p2p;
    p2p.SetDeviceAttribute ("DataRate", StringValue ("5Mbps"));
    p2p.SetChannelAttribute ("Delay", StringValue ("2ms"));

    NetDeviceContainer devices1 = p2p.Install (senders.Get (0), routers.Get (0));
    NetDeviceContainer devices2 = p2p.Install (senders.Get (1), routers.Get (0));
    NetDeviceContainer devices3 = p2p.Install (senders.Get (2), routers.Get (1));
    NetDeviceContainer devices4 = p2p.Install (senders.Get (3), routers.Get (1));

    // Bottleneck link (between routers)
    PointToPointHelper bottleneck;
    bottleneck.SetDeviceAttribute ("DataRate", StringValue ("1Mbps"));
    bottleneck.SetChannelAttribute ("Delay", StringValue ("10ms"));

    NetDeviceContainer routerDevices = bottleneck.Install (routers.Get (0), routers.Get (1));

    // Links between routers and receivers
    NetDeviceContainer devices5 = p2p.Install (routers.Get (0), receivers.Get (0));
    NetDeviceContainer devices6 = p2p.Install (routers.Get (0), receivers.Get (1));
    NetDeviceContainer devices7 = p2p.Install (routers.Get (1), receivers.Get (2));
    NetDeviceContainer devices8 = p2p.Install (routers.Get (1), receivers.Get (3));

    // Install Internet Stack
    InternetStackHelper stack;
    stack.Install (senders);
    stack.Install (receivers);
    stack.Install (routers);

```

```

// Assign IP addresses
Ipv4AddressHelper address;
address.SetBase ("10.1.1.0", "255.255.255.0");
Ipv4InterfaceContainer senderInterfaces1 = address.Assign (devices1);

address.SetBase ("10.1.2.0", "255.255.255.0");
Ipv4InterfaceContainer senderInterfaces2 = address.Assign (devices2);

address.SetBase ("10.1.3.0", "255.255.255.0");
Ipv4InterfaceContainer senderInterfaces3 = address.Assign (devices3);

address.SetBase ("10.1.4.0", "255.255.255.0");
Ipv4InterfaceContainer senderInterfaces4 = address.Assign (devices4);

address.SetBase ("10.1.5.0", "255.255.255.0");
Ipv4InterfaceContainer routerInterfaces = address.Assign (routerDevices);

address.SetBase ("10.1.6.0", "255.255.255.0");
Ipv4InterfaceContainer receiverInterfaces1 = address.Assign (devices5);

address.SetBase ("10.1.7.0", "255.255.255.0");
Ipv4InterfaceContainer receiverInterfaces2 = address.Assign (devices6);

address.SetBase ("10.1.8.0", "255.255.255.0");
Ipv4InterfaceContainer receiverInterfaces3 = address.Assign (devices7);

address.SetBase ("10.1.9.0", "255.255.255.0");
Ipv4InterfaceContainer receiverInterfaces4 = address.Assign (devices8);

Ipv4GlobalRoutingHelper::PopulateRoutingTables ();

// UDP traffic generation
uint16_t port = 9; // UDP port

OnOffHelper onoff1 ("ns3::UdpSocketFactory", InetSocketAddress
(receiverInterfaces1.GetAddress (0), port));
onoff1.SetConstantRate (DataRate (std::to_string (trafficRate) + "Mbps"));
onoff1.SetAttribute ("PacketSize", UintegerValue (1024));

OnOffHelper onoff2 ("ns3::UdpSocketFactory", InetSocketAddress
(receiverInterfaces2.GetAddress (0), port));
onoff2.SetConstantRate (DataRate (std::to_string (trafficRate) + "Mbps"));
onoff2.SetAttribute ("PacketSize", UintegerValue (1024));

OnOffHelper onoff3 ("ns3::UdpSocketFactory", InetSocketAddress
(receiverInterfaces3.GetAddress (0), port));
onoff3.SetConstantRate (DataRate (std::to_string (trafficRate) + "Mbps"));
onoff3.SetAttribute ("PacketSize", UintegerValue (1024));

```



```

OnOffHelper onoff4 ("ns3::UdpSocketFactory", InetSocketAddress
(receiverInterfaces4.GetAddress (0), port));
onoff4.SetConstantRate (DataRate (std::to_string (trafficRate) + "Mbps"));
onoff4.SetAttribute ("PacketSize", UIntegerValue (1024));

ApplicationContainer apps;
apps.Add (onoff1.Install (senders.Get (0)));
apps.Add (onoff2.Install (senders.Get (1)));
apps.Add (onoff3.Install (senders.Get (2)));
apps.Add (onoff4.Install (senders.Get (3)));

apps.Start (Seconds (1.0));
apps.Stop (Seconds (5.0));

// Packet sinks at receivers
PacketSinkHelper sink ("ns3::UdpSocketFactory", InetSocketAddress (Ipv4Address::GetAny
()), port));
ApplicationContainer sinkApps;
sinkApps.Add (sink.Install (receivers.Get (0)));
sinkApps.Add (sink.Install (receivers.Get (1)));
sinkApps.Add (sink.Install (receivers.Get (2)));
sinkApps.Add (sink.Install (receivers.Get (3)));

sinkApps.Start (Seconds (0.0));
sinkApps.Stop (Seconds (6.0));

// Flow Monitor
FlowMonitorHelper flowmon;
Ptr<FlowMonitor> monitor = flowmon.InstallAll ();

Simulator::Stop (Seconds (6.0));
Simulator::Run ();

monitor->CheckForLostPackets ();

Ptr<Ipv4FlowClassifier> classifier = DynamicCast<Ipv4FlowClassifier> (flowmon.GetClassifier
());
std::map<FlowId, FlowMonitor::FlowStats> stats = monitor->GetFlowStats ();

if (stats.size() == 0)
{
    std::cout << "No flows detected!" << std::endl;
}

for (auto it = stats.begin (); it != stats.end (); ++it)
{
    Ipv4FlowClassifier::FiveTuple t = classifier->FindFlow (it->first);
    if (t.destinationPort == port)
    {

```

```

        double throughput = (it->second.rxBytes * 8.0) / (5.0 * 1000000.0); // Convert to Mbps
        std::cout << "Traffic Rate = " << trafficRate
            << " Mbps --> Flow from " << t.sourceAddress
            << " to " << t.destinationAddress
            << " --> Throughput = " << throughput
            << " Mbps" << std::endl;
    }
}

Simulator::Destroy ();
}

int main (int argc, char *argv[])
{
    NS_LOG_UNCOND ("Starting Bottleneck Analysis...");

    RunSimulation (1); // 1 Mbps
    RunSimulation (2); // 2 Mbps
    RunSimulation (5); // 5 Mbps

    return 0;
}

```

The screenshot shows a C++ IDE with three tabs: bandwidth_analysis.cc, buffer_management.cc, and bottleneck_analysis.cc. The active tab, bottleneck_analysis.cc, contains the following code:

```

1 #include "ns3/core-module.h"
2 #include "ns3/network-module.h"
3 #include "ns3/internet-module.h"
4 #include "ns3/point-to-point-module.h"
5 #include "ns3/applications-module.h"
6 #include "ns3/flow-monitor-module.h"
7
8 using namespace ns3;
9
10 NS_LOG_COMPONENT_DEFINE ("BottleneckAnalysisExample");
11
12 void RunSimulation (uint32_t trafficRate)
13 {
14     std::cout << "\nRunning Simulation for Traffic Rate = " << trafficRate << " Mbps...\n";
15
16     NodeContainer senders, receivers, routers;
17     senders.Create (4);
18     receivers.Create (4);
19     routers.Create (2);
20
21     // Set up point-to-point links between senders and routers
22     PointToPointHelper p2p;
23     p2p.SetDeviceAttribute ("DataRate", StringValue ("5Mbps"));
24     p2p.SetChannelAttribute ("Delay", StringValue ("2ms"));
25
26     NetDeviceContainer devices1 = p2p.Install (senders.Get (0), routers.Get (0));
27     NetDeviceContainer devices2 = p2p.Install (senders.Get (1), routers.Get (0));
28     NetDeviceContainer devices3 = p2p.Install (senders.Get (2), routers.Get (1));
29     NetDeviceContainer devices4 = p2p.Install (senders.Get (3), routers.Get (1));
30
31     // Bottleneck link (between routers)

```

The IDE interface includes a top bar with 'Open', 'Save', and window controls. The bottom status bar shows 'C++', 'Tab Width: 8', 'Ln 14, Col 89', and 'INS'.

```

[0/2] Re-checking globbed directories...
[2/2] Linking CXX executable ../build/scratch/ns3.43-bottleneck_analysis-default
Starting Bottleneck Analysis...

Running Simulation for Traffic Rate = 1 Mbps...
No flows detected!

Running Simulation for Traffic Rate = 2 Mbps...
No flows detected!

Running Simulation for Traffic Rate = 5 Mbps...
No flows detected!
ubuntu@ubuntu-virtual-machine:/ns-allinone-3.43/ns-3.43$ ./ns3 run scratch/bottleneck_analysis
[0/2] Re-checking globbed directories...
[2/2] Linking CXX executable ../build/scratch/ns3.43-bottleneck_analysis-default
Starting Bottleneck Analysis...

Running Simulation for Traffic Rate = 1 Mbps...
Traffic Rate = 1 Mbps --> Flow from 10.1.1.1 to 10.1.6.1 --> Throughput = 0.821402 Mbps
Traffic Rate = 1 Mbps --> Flow from 10.1.2.1 to 10.1.7.1 --> Throughput = 0.821402 Mbps
Traffic Rate = 1 Mbps --> Flow from 10.1.3.1 to 10.1.8.1 --> Throughput = 0.821402 Mbps
Traffic Rate = 1 Mbps --> Flow from 10.1.4.1 to 10.1.9.1 --> Throughput = 0.821402 Mbps

Running Simulation for Traffic Rate = 2 Mbps...
Traffic Rate = 2 Mbps --> Flow from 10.1.1.1 to 10.1.6.1 --> Throughput = 1.6428 Mbps
Traffic Rate = 2 Mbps --> Flow from 10.1.2.1 to 10.1.7.1 --> Throughput = 1.6428 Mbps
Traffic Rate = 2 Mbps --> Flow from 10.1.3.1 to 10.1.8.1 --> Throughput = 1.6428 Mbps
Traffic Rate = 2 Mbps --> Flow from 10.1.4.1 to 10.1.9.1 --> Throughput = 1.6428 Mbps

Running Simulation for Traffic Rate = 5 Mbps...
Traffic Rate = 5 Mbps --> Flow from 10.1.1.1 to 10.1.6.1 --> Throughput = 4.10869 Mbps
Traffic Rate = 5 Mbps --> Flow from 10.1.2.1 to 10.1.7.1 --> Throughput = 4.10869 Mbps
Traffic Rate = 5 Mbps --> Flow from 10.1.3.1 to 10.1.8.1 --> Throughput = 4.10869 Mbps
Traffic Rate = 5 Mbps --> Flow from 10.1.4.1 to 10.1.9.1 --> Throughput = 4.10869 Mbps

```

Task5

Data_loss.cc

```

#include "ns3/core-module.h"
#include "ns3/network-module.h"
#include "ns3/internet-module.h"
#include "ns3/yans-wifi-helper.h"
#include "ns3/mobility-module.h"
#include "ns3/applications-module.h"
#include "ns3/flow-monitor-module.h"
#include "ns3/wifi-module.h" // <-- Add this line

```

```
using namespace ns3;
```

```
NS_LOG_COMPONENT_DEFINE ("DataLossSimulation");
```

```
void ConfigureMobility (NodeContainer &staNodes, NodeContainer &apNode, double distance)
{
```

```

    MobilityHelper mobility;
    mobility.SetPositionAllocator ("ns3::GridPositionAllocator",
                                  "MinX", DoubleValue (0.0),
                                  "MinY", DoubleValue (0.0),
                                  "DeltaX", DoubleValue (distance),
                                  "DeltaY", DoubleValue (distance),
                                  "GridWidth", UIntegerValue (1),
                                  "LayoutType", StringValue ("RowFirst"));

```

```

mobility.SetMobilityModel ("ns3::ConstantPositionMobilityModel");
mobility.Install (staNodes);

mobility.SetPositionAllocator ("ns3::GridPositionAllocator",
                               "MinX", DoubleValue (0.0),
                               "MinY", DoubleValue (0.0),
                               "DeltaX", DoubleValue (0.0),
                               "DeltaY", DoubleValue (0.0),
                               "GridWidth", UIntegerValue (1),
                               "LayoutType", StringValue ("RowFirst"));
mobility.Install (apNode);
}

int main (int argc, char *argv[])
{
    double distance = 10.0;
    CommandLine cmd;
    cmd.AddValue ("distance", "Distance between AP and STAs", distance);
    cmd.Parse (argc, argv);

    NodeContainer wifiStaNodes;
    wifiStaNodes.Create (3);
    NodeContainer wifiApNode;
    wifiApNode.Create (1);

    YansWifiChannelHelper channel = YansWifiChannelHelper::Default ();
    YansWifiPhyHelper phy;
    phy.SetChannel (channel.Create ());

    WifiHelper wifi;
    wifi.SetRemoteStationManager ("ns3::IdealWifiManager");

    WifiMacHelper mac;
    Ssid ssid = Ssid ("ns-3-ssid");

    mac.SetType ("ns3::StaWifiMac",
                "Ssid", SsidValue (ssid),
                "ActiveProbing", BooleanValue (false));
    NetDeviceContainer staDevices;
    staDevices = wifi.Install (phy, mac, wifiStaNodes);

    mac.SetType ("ns3::ApWifiMac",
                "Ssid", SsidValue (ssid));
    NetDeviceContainer apDevice;
    apDevice = wifi.Install (phy, mac, wifiApNode);

    InternetStackHelper stack;
    stack.Install (wifiApNode);

```

```

stack.Install (wifiStaNodes);
Ipv4AddressHelper address;
address.SetBase ("10.1.3.0", "255.255.255.0");
Ipv4InterfaceContainer staInterfaces;
staInterfaces = address.Assign (staDevices);
address.Assign (apDevice);
// Mobility
ConfigureMobility (wifiStaNodes, wifiApNode, distance);
// UDP Traffic from AP to STAs
uint16_t port = 9;
UdpServerHelper server (port);
ApplicationContainer serverApps;
for (uint32_t i = 0; i < wifiStaNodes.GetN (); ++i)
{
    serverApps.Add (server.Install (wifiStaNodes.Get (i)));
}
serverApps.Start (Seconds (1.0));
serverApps.Stop (Seconds (10.0));
UdpClientHelper client (staInterfaces.GetAddress (0), port);
client.SetAttribute ("MaxPackets", UintegerValue (10000));
client.SetAttribute ("Interval", TimeValue (Seconds (0.01)));
client.SetAttribute ("PacketSize", UintegerValue (1024));
ApplicationContainer clientApps;
clientApps.Add (client.Install (wifiApNode.Get (0)));
clientApps.Start (Seconds (2.0));
clientApps.Stop (Seconds (10.0));
// Interference traffic (Station-to-Station)
UdpClientHelper interClient (staInterfaces.GetAddress (1), port);
interClient.SetAttribute ("MaxPackets", UintegerValue (10000));
interClient.SetAttribute ("Interval", TimeValue (Seconds (0.01)));
interClient.SetAttribute ("PacketSize", UintegerValue (1024));
ApplicationContainer interApps;
interApps.Add (interClient.Install (wifiStaNodes.Get (2)));
interApps.Start (Seconds (2.5));
interApps.Stop (Seconds (10.0));
// Flow Monitor
FlowMonitorHelper flowmon;
Ptr<FlowMonitor> monitor = flowmon.InstallAll ();
Simulator::Stop (Seconds (11.0));
Simulator::Run ();
monitor->CheckForLostPackets ();
Ptr<Ipv4FlowClassifier> classifier = DynamicCast<Ipv4FlowClassifier>
(flowmon.GetClassifier ());
std::map<FlowId, FlowMonitor::FlowStats> stats = monitor->GetFlowStats ();

for (auto iter = stats.begin (); iter != stats.end (); ++iter)
{
    Ipv4FlowClassifier::FiveTuple t = classifier->FindFlow (iter->first);

```

```

        std::cout << "Flow " << iter->first << " (" << t.sourceAddress << " -> " <<
t.destinationAddress << ")\n";
        std::cout << " Tx Packets: " << iter->second.txPackets << "\n";
        std::cout << " Rx Packets: " << iter->second.rxPackets << "\n";
        std::cout << " Lost Packets: " << (iter->second.txPackets - iter->second.rxPackets) <<
"\n";
        std::cout << " Throughput: " << (iter->second.rxBytes * 8.0 / (iter-
>second.timeLastRxPacket.GetSeconds() - iter->second.timeFirstTxPacket.GetSeconds())) /
1024 / 1024 << " Mbps\n";
    }
    Simulator::Destroy ();
    return 0;
}

```

```

data_loss.cc
~/ns-allinone-3.43/ns-3.43/scratch

bandwidth_analysis.cc  x  buffer_management.cc  x  bottleneck_analysis.cc  x  data_loss.cc  x

1 #include "ns3/core-module.h"
2 #include "ns3/network-module.h"
3 #include "ns3/internet-module.h"
4 #include "ns3/yans-wifi-helper.h"
5 #include "ns3/mobility-module.h"
6 #include "ns3/applications-module.h"
7 #include "ns3/flow-monitor-module.h"
8 #include "ns3/wifi-module.h" // <-- Add this line
9
10 using namespace ns3;
11
12 NS_LOG_COMPONENT_DEFINE ("DataLossSimulation");
13
14 void ConfigureMobility (NodeContainer &staNodes, NodeContainer &apNode, double distance)
15 {
16     MobilityHelper mobility;
17     mobility.SetPositionAllocator ("ns3::GridPositionAllocator",
18                                   "MinX", DoubleValue (0.0),
19                                   "MinY", DoubleValue (0.0),
20                                   "DeltaX", DoubleValue (distance),
21                                   "DeltaY", DoubleValue (distance),
22                                   "GridWidth", UIntegerValue (1),
23                                   "LayoutType", StringValue ("RowFirst"));
24
25     mobility.SetMobilityModel ("ns3::ConstantPositionMobilityModel");
26     mobility.Install (staNodes);
27
28     mobility.SetPositionAllocator ("ns3::GridPositionAllocator",
29                                   "MinX", DoubleValue (0.0),
30                                   "MinY", DoubleValue (0.0),
31                                   "DeltaX", DoubleValue (0.0),
32                                   "DeltaY", DoubleValue (0.0));
33 }

```

C++ Tab Width: 8 Ln 15, Col 2 INS

```

[2/2] Linking CXX executable ../build/scratch/ns3.43-data_loss-default
Flow 1 (10.1.3.4 -> 10.1.3.1)
Tx Packets: 800
Rx Packets: 800
Lost Packets: 0
Throughput: 0.803609 Mbps
Flow 2 (10.1.3.3 -> 10.1.3.2)
Tx Packets: 750
Rx Packets: 750
Lost Packets: 0
Throughput: 0.803635 Mbps
ubuntu@ubuntu-virtual-machine: ~/ns-allinone-3.43/ns-3.43$ ./ns3 run "scratch/data_loss --distance=50"
[0/2] Re-checking glopped directories...
ninja: no work to do.
Flow 1 (10.1.3.4 -> 10.1.3.1)
Tx Packets: 800
Rx Packets: 800
Lost Packets: 0
Throughput: 0.803609 Mbps
Flow 2 (10.1.3.3 -> 10.1.3.2)
Tx Packets: 750
Rx Packets: 0
Lost Packets: 750
Throughput: -0 Mbps
ubuntu@ubuntu-virtual-machine: ~/ns-allinone-3.43/ns-3.43$ ./ns3 run "scratch/data_loss --distance=100"
[0/2] Re-checking glopped directories...
ninja: no work to do.
Flow 1 (10.1.3.4 -> 10.1.3.1)
Tx Packets: 800
Rx Packets: 800
Lost Packets: 0
Throughput: 0.803609 Mbps
Flow 2 (10.1.3.3 -> 10.1.3.2)
Tx Packets: 750
Rx Packets: 0
Lost Packets: 750
Throughput: -0 Mbps

```

Flow1: Without Interference

Flow2: With Interference.